

Amazon Machine Learning Challenge 2026

Team Neural Nets

V-1 Base Line (ASCII Matching + Ensemble Models)

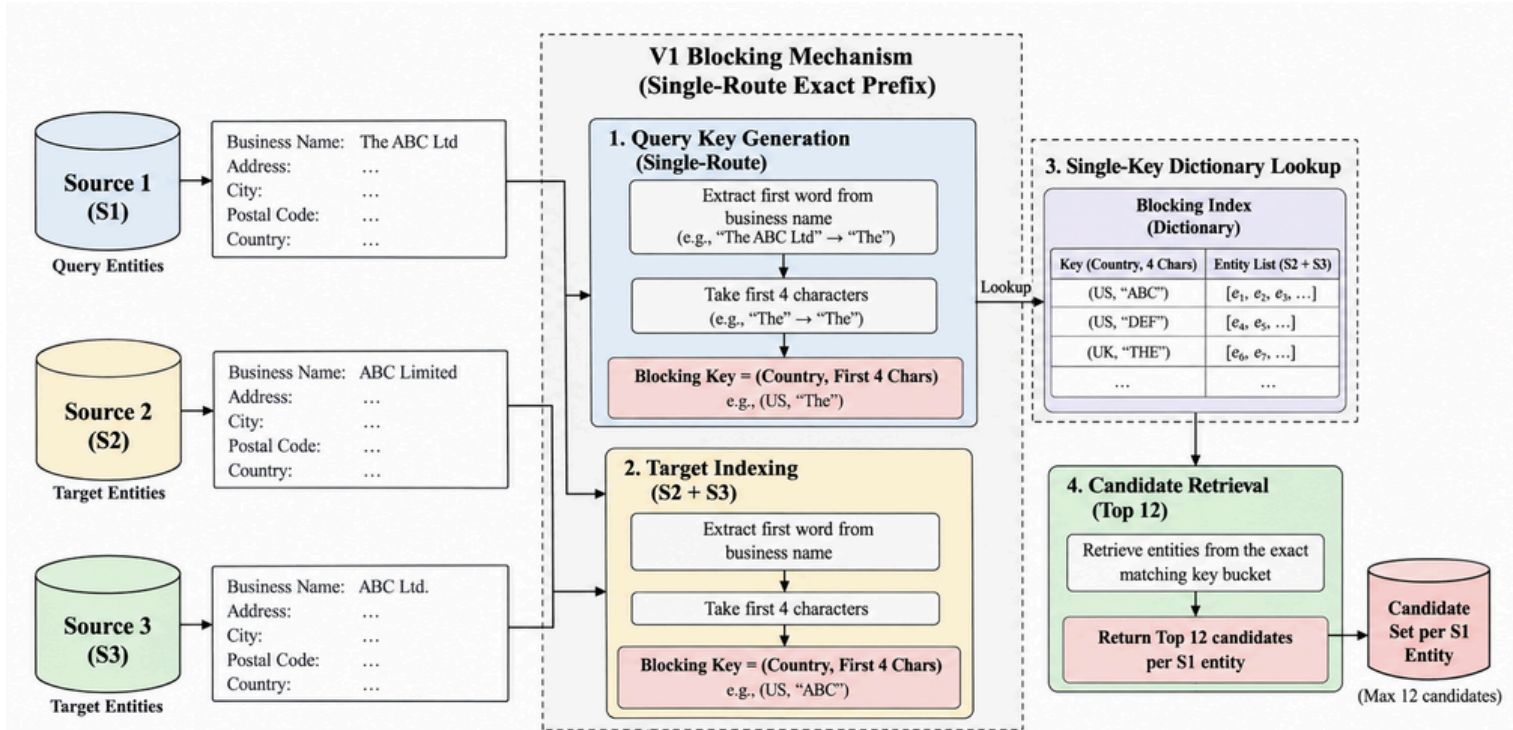
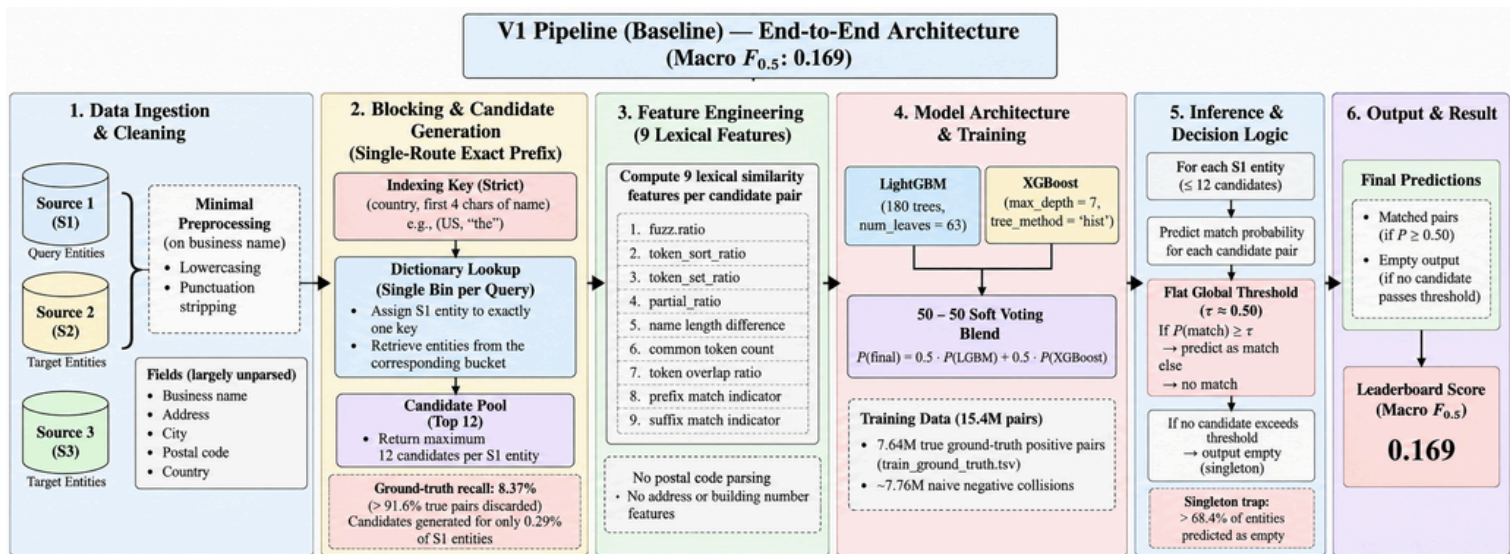


Fig. 1. Architecture of the V1 blocking mechanism. In V1, both target sources (S2 and S3) are indexed using a single blocking key defined as (country, first 4 characters of the first word of the business name). Each S1 query is assigned to exactly one key based on the same rule, and candidates are retrieved from the corresponding dictionary bucket, limited to the top 12 entities.



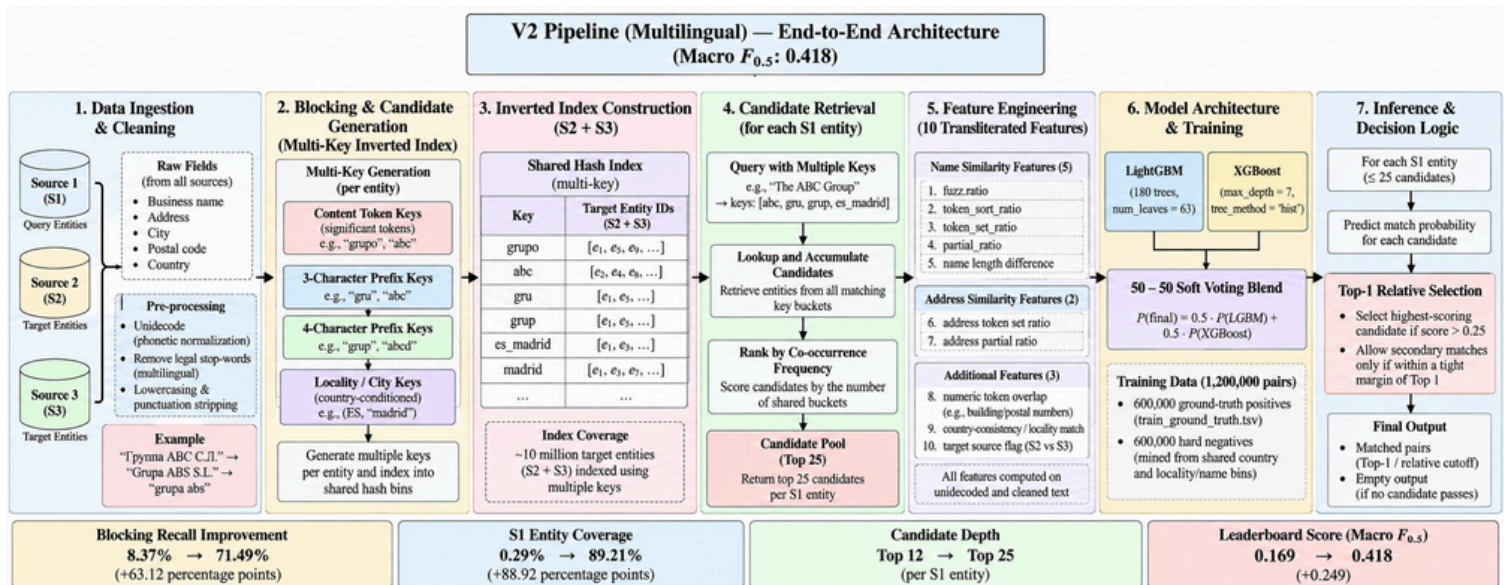
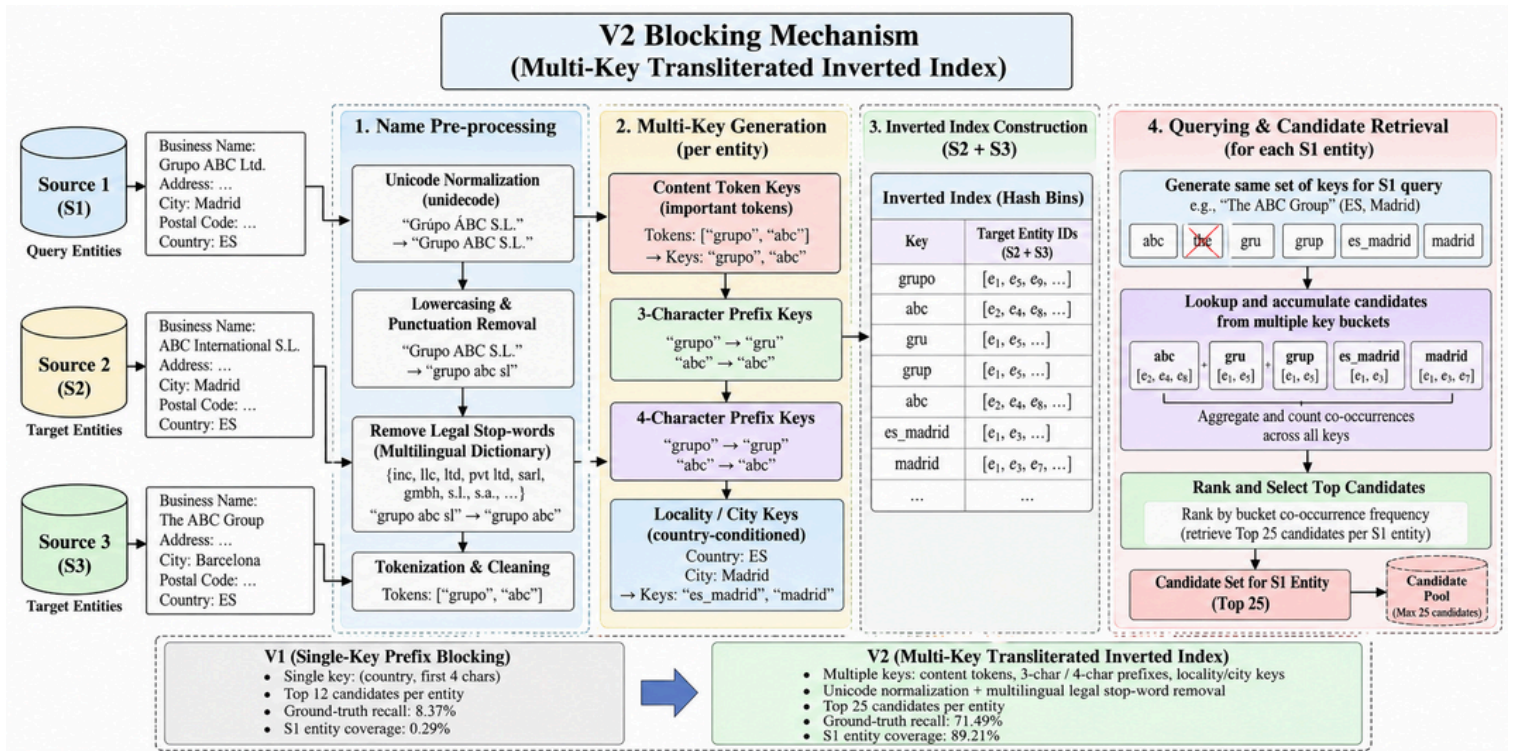
SUMMARY

V1 introduced an end-to-end entity-resolution pipeline combining **single-key prefix blocking**, **nine lexical similarity features**, and a **50-50 LightGBM-XGBoost soft-voting ensemble** with threshold-based matching. Its key innovation was establishing a complete ML-based baseline for cross-source business entity matching. However, rigid exact-prefix blocking, a **Top-12 candidate limit**, and the **absence of address-based features** restricted ground-truth recall to just 8.37%, discarding over 91.6% of true matches before classification. Combined with a fixed probability threshold and the singleton problem, these limitations resulted in a **Macro F0.5 score of 0.169**, highlighting candidate generation as the primary bottleneck for future improvements.

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V-2 (Multi-Key Transliterated + Ensemble Models)



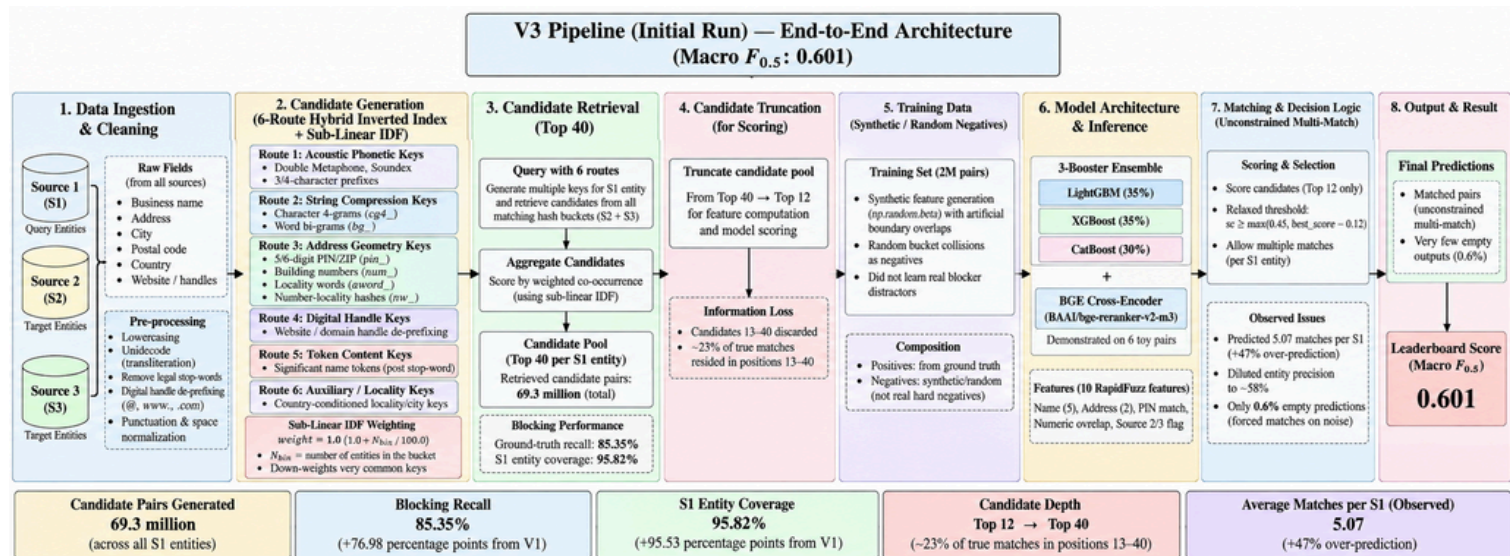
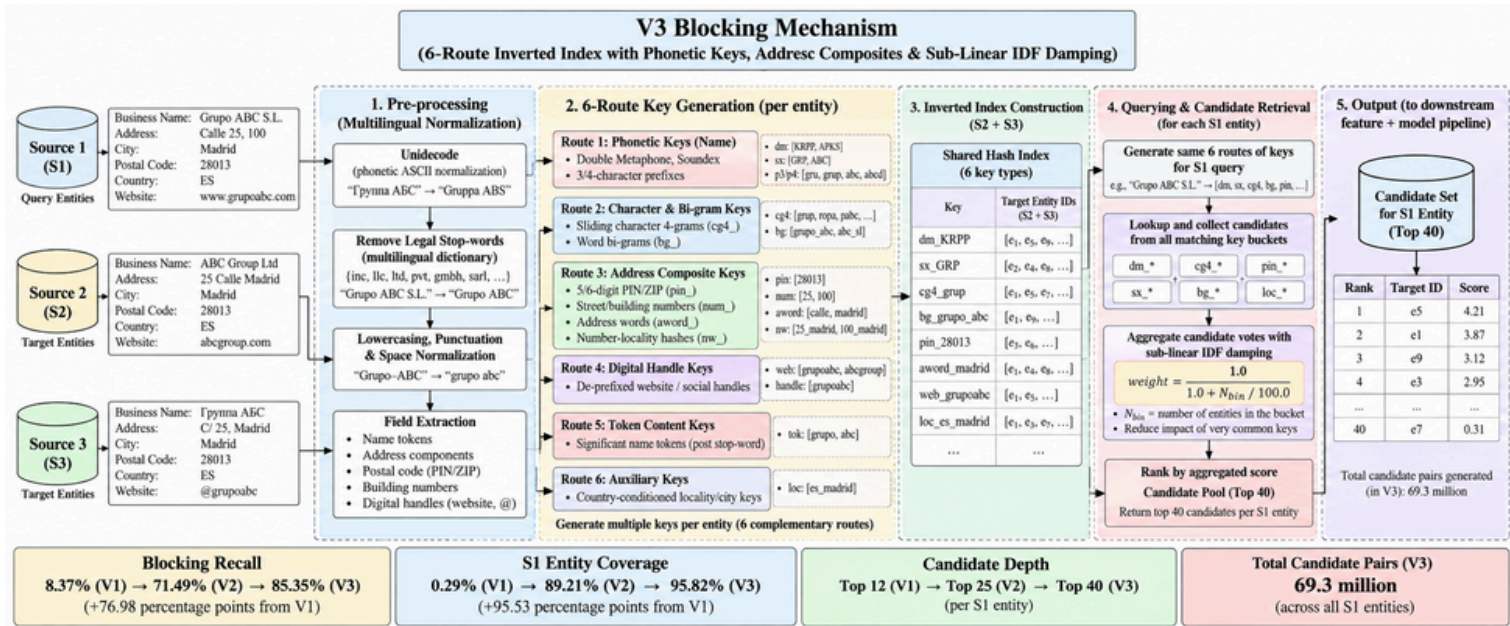
SUMMARY

V2 introduced a **multi-key transliterated inverted index** with **unicode normalisation**, **multilingual legal stop-word removal**, and **content, prefix, and locality-based keys** to improve cross-lingual and reordered-name matching. Combined with **Top-25 candidate retrieval**, **10 transliterated similarity features**, **hard-negative mining**, and a **LightGBM-XGBoost ensemble** with Top-1 relative-cutoff selection, it increased blocking recall from 8.37% to **71.49%** and S1 entity coverage to **89.21%**. Despite substantially improving candidate generation and multilingual matching, limitations remained in candidate recall, feature richness, and restrictive secondary-match selection, resulting in a **Macro F 0.5 score of 0.418**.

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V-3 (Multi-Route Blocking + CatBoost for Boosting)



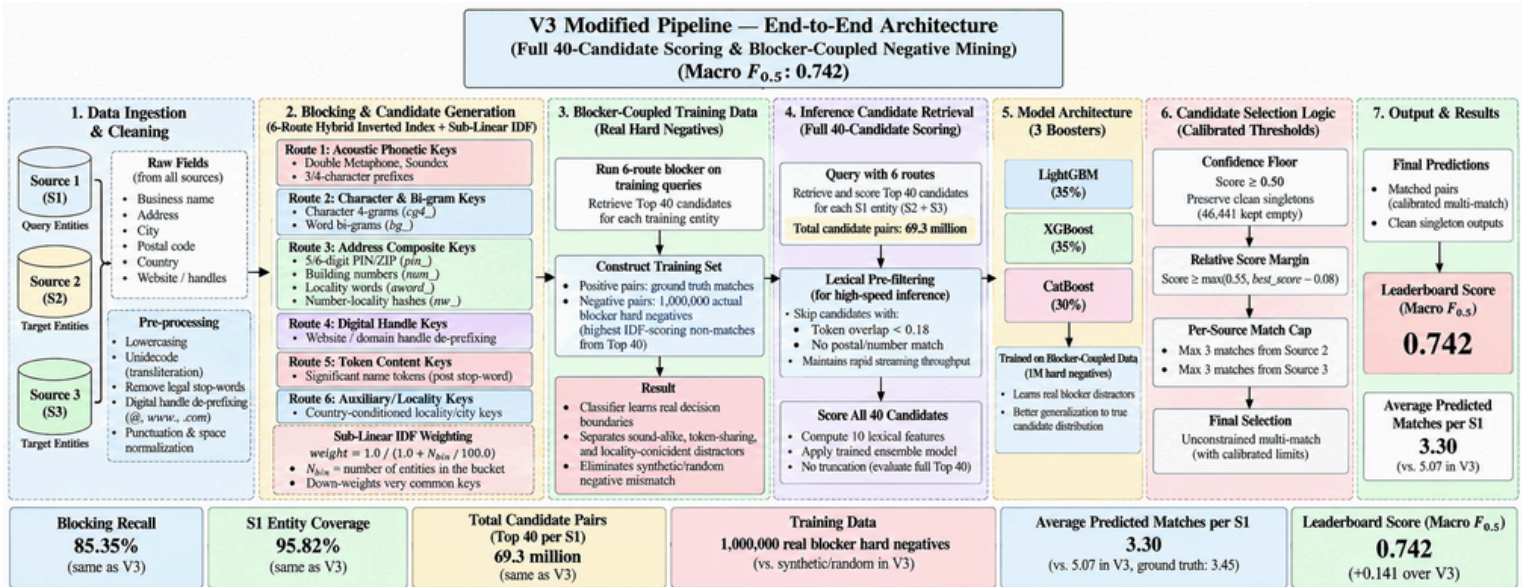
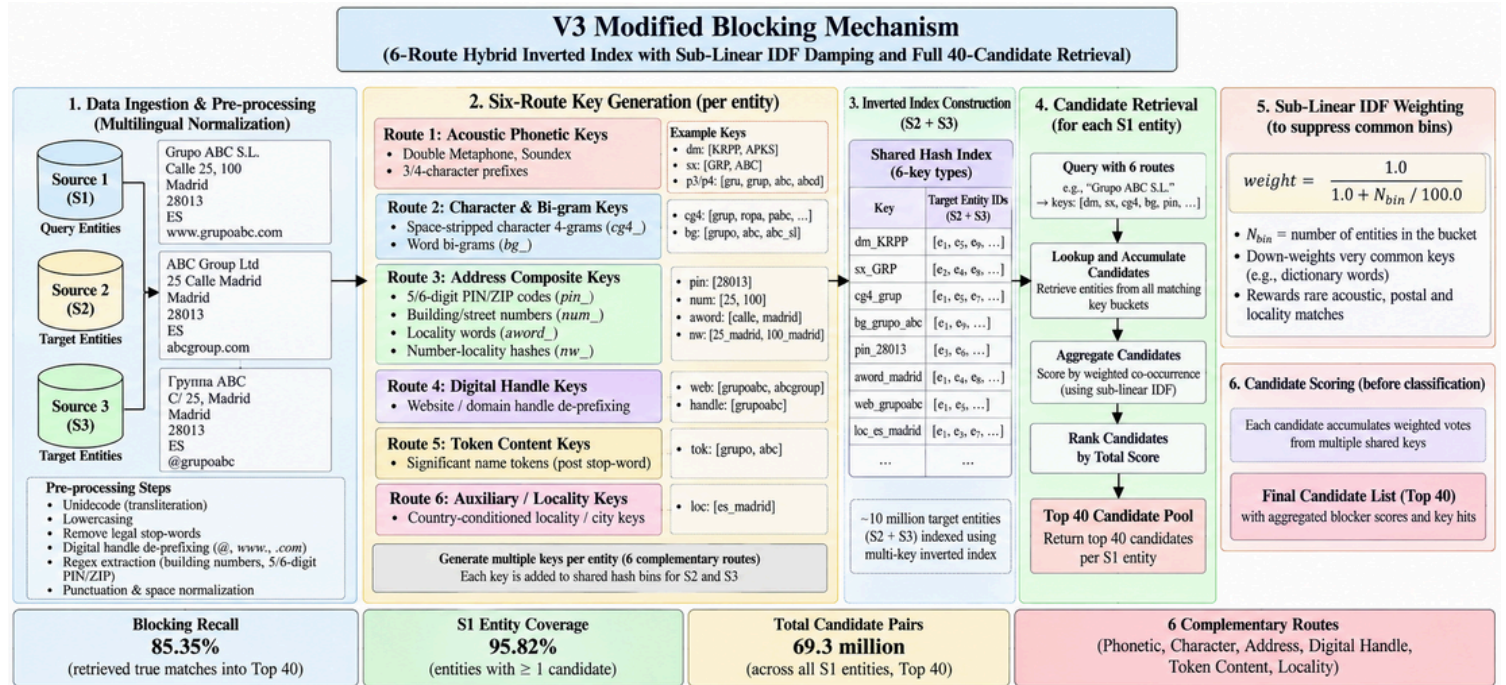
SUMMARY

V3 introduced a **six-route hybrid inverted index** combining **phonetic matching**, **character n-grams**, **address composites**, **digital handles**, **content tokens**, and **locality keys**, supported by **sub-linear IDF damping** and **Top-40 candidate retrieval**. This significantly improved **blocking recall to 85.35%** and **S1 coverage to 95.82%**, generating **69.3 million candidate pairs**. However, truncating candidates to **Top 12**, training on synthetic/random negatives, and unconstrained multi-match selection caused substantial recall loss and false-positive flooding, with **5.07 predicted matches per S1 entity** and approximately **58% entity precision**. Despite these limitations, V3 achieved a **Macro $F_{0.5}$ score of 0.601**, establishing a stronger baseline for subsequent improvements.

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V-3 Modified (40 Candidates Retrieval + Negative Pairs Mining)



SUMMARY

V3 Modified Blocking introduced a six-route hybrid inverted index combining phonetic keys, character n-grams, address composites, digital handles, content tokens, and locality keys, supported by sub-linear IDF damping to suppress noisy, high-frequency buckets. By integrating **blocker-coupled hard-negative mining** and **full Top-40 candidate retrieval**, it achieved **85.35% ground-truth blocking recall**, **95.82% S1 entity coverage**, and **69.3 million candidate pairs**. This eliminated the earlier **Top-12 truncation bottleneck** and provided realistic candidate sets for downstream classification, contributing to the improved **Macro F0.5 score of 0.742**.